ENGINE [SKYACTIV-D 1.5]

id0100000002q1

Special Features

- For SKYACTIV-D 1.5, the following have been adopted to lower fuel consumption.
  - — Low compression ratio
    - Combustion efficiency by lower compression ration (14.8)
  - — Weight reductions
    - Aluminum alloy cylinder block adopted
    - Hard-plastic intake manifold adopted
    - Exhaust manifold integrated cylinder head adopted
  - — Improvement of mechanical resistance loss
    - Optimized oil passage
    - Optimized oil pump shape
    - Optimized engine coolant passage
    - Optimized water pump impeller shape
  - — Weight reduction and mechanical resistance loss improvements
    - Piston shape optimized
    - Narrowed down crankshaft journal
  - — Heat loss reduction
    - Water jacket spacer adopted
  - — Cooling loss improvement
    - Coolant control valve adopted
- Glow control has been adopted to improve engine startability and diesel particulate filter regeneration performance.
  - — Glow control module adopted
- Engine hydraulic pressure switching control has been adopted reducing engine load.
  - — Engine oil solenoid valve adopted
- An exhaust gas recirculation (EGR) system has been adopted which achieves cleaner exhaust emissions and improved fuel efficiency.
- Boost control has been adopted for improved vehicle acceleration performance, fuel economy, and environmental performance.
  - — Variable geometry turbocharger adopted
  - — Speed sensor component (variable geometry turbocharger) adopted
- Generator output control adopted, fuel economy/idling stability improved.
  - — Current sensor adopted
- LIN communication has been adopted to the current sensor and DC-DC converter (Without i-ELOOP) for simplified wiring harnesses.
- i-stop control has been adopted for improved fuel efficiency, reduced exhaust gas emissions, and reduced idling noise.
- DC-DC converter control has been adopted for improved power supply stability. (Without i-ELOOP)
  - — DC-DC converter adopted
- With the adoption of the i-ELOOP, charging efficiency during deceleration is improved. (See i-ELOOP [i-ELOOP].) (With i-ELOOP)
  - — DC-DC converter (i-ELOOP) adopted

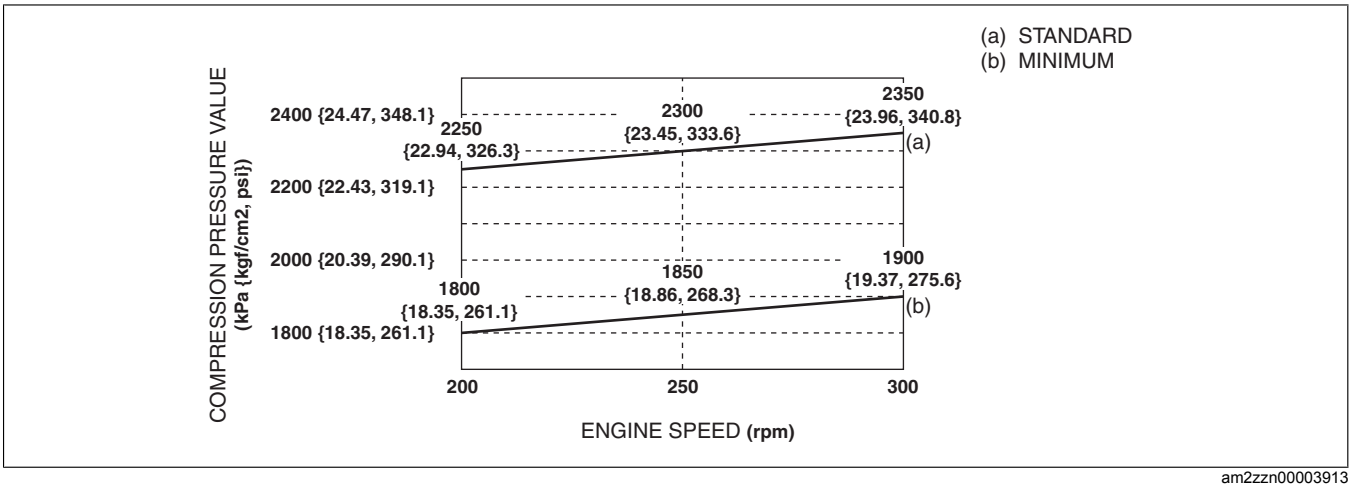
Specification

Item				Specification
MECHANICAL				
Type				Diesel, 4-cycle
Cylinder arrangement and number				In-line, 4-cylinder
Combustion chamber				Direct injection
Valve system				DOHC, timing chain driven, 16 valves
Displacement			(ml {cc})	1,498 {1,498}
Bore × stroke			(mm {in})	76.0 × 82.6 {2.99 × 3.25}
Compression ratio				14.8:1
Valve timing	IN	Open	(°)	BTDC 3
		Closed	(°)	ABDC 25
	EX	Open	(°)	BBDC 40
		Closed	(°)	ATDC 8
LUBRICATION SYSTEM				
Type				Force-fed type

Item			Specification
Oil pressure (reference value) [Oil temperature: 80—90 °C {176—194 °F}, Water temperature: 80—90 °C {176—194 °F}]		(kPa {kgf/cm^2, psi} [rpm])	Low oil pressure: 110—175 {1.13—1.78, 16.0—25.3}[1,500] High oil pressure: 300—430 {3.06—4.38, 43.6—62.3}[3,000]
Oil pump	Type		Trochoid gear type
Oil filter	Type		Full-flow, paper element
	Bypass pressure	(kPa {kgf/cm^2, psi})	Type A: 78—118 {0.80—1.20, 12.0—17.1} Type B: 225—265 {2.30—2.70, 32.7—38.4}
Oil capacity (approx. quantity)	Oil replacement	(L {US qt, Imp qt})	4.7 {5.0, 4.1}
	Oil and oil filter replacement	(L {US qt, Imp qt})	5.1 {5.4, 4.5}
	Total (dry engine)	(L {US qt, Imp qt})	5.8 {6.1, 5.1}
COOLING SYSTEM			
Type			Water-cooled, Electromotive
Coolant capacity	Engine	(L {US qt, Imp qt})	MTX: 6.6 {7.0, 5.8} ATX (Israel): 6.7 {7.1, 5.9} ATX (Except Israel): 6.8 {7.2, 6.0}
	Water-cooled charge air cooler	(L {US qt, Imp qt})	2.0 {2.1, 1.8}
Water pump	Type		Centrifugal, V-ribbed belt-driven
Electric water pump	Type		Electromotive
Thermostat (Built into the coolant control valve)	Type		Wax type
	Opening temperature	(°C {°F})	101—105 {214—221}
	Full-open temperature	(°C {°F})	116 {241}
	Full-open lift	(mm {in})	3 {0.1} or more
Radiator	Type		Corrugated fin type
Cooling system cap	Valve opening pressure	(kPa {kgf/cm^2, psi})	93.2—122.6 {0.951—1.250, 13.6—17.7}
Water-cooled charge air cooler reservoir tank cap	Valve opening pressure	(kPa {kgf/cm^2, psi})	24—36 {0.25—0.36, 3.5—5.2}
Cooling fan	Type		Electric type
	Number of blades		7
	Outer diameter	(mm {in})	340 {13.4}
	Cooling fan motor output	(W)	240
FUEL SYSTEM			
Supply pump	Type		Electric control
Fuel injector	Type		Electric control
Fuel tank	Capacity	(L {US gal, Imp gal})	44.0
Fuel type	Type		EN590 or the equivalent
CHARGING SYSTEM			
Battery	Voltage	(V)	12
	Type		With i-ELOOP: Q-85 Without i-ELOOP: S-95
Generator	Output	(V-A)	12-150
STARTING SYSTEM			
Starter	Type		Coaxial reduction
	Output	(kW)	1.7
CONTROL SYSTEM			
i-stop OFF switch			ON/OFF
Neutral switch			ON/OFF
CPP switch			ON/OFF
MAF sensor			Hot film
IAT sensor			Thermistor
MAP sensor			Semiconductor strain gauge
APP sensor			Electromagnetic inductiont
CKP sensor			MR (Magnetic Resistance) element
CMP sensor			MR (Magnetic Resistance) element
ECT sensor			Thermistor
BARO sensor			Piezoelectric element
Fuel pressure sensor			Piezoelectric element
Coolant control valve			Contact-type rotary
Current sensor			Shunt resistance, Thermistor

Item	Specification
A/F sensor	zirconium element
Boost air temperature sensor	Thermistor
Fuel temperature sensor	Thermistor
Exhaust gas temperature sensor	Thermistor
Exhaust gas pressure sensor No.1	Semiconductor strain gauge
Exhaust gas pressure sensor No.2	Piezoelectric element
Engine oil temperature sensor	Thermistor
Engine oil pressure sensor	Semiconductor strain gauge
Intake shutter valve position sensor	Hall element
Exhaust shutter valve position sensor	Hall element
HP-EGR control valve position sensor	Hall element
LP-EGR control valve position sensor	Hall element
Speed sensor assy (turbocharger with variable turbine geometry)	MR (Magnetic Resistance) element
Actuator position sensor (turbocharger with variable turbine geometry)	Hall element
Clutch stroke sensor	Hall element
Power brake unit vacuum sensor	Piezoelectric element

Compression pressure



am2zzn00003913

Engine oil [U.K. specs.]

Recommended engine oil		Alternative engine oil
Mazda Original Oil Supra DPF 0W-30	Mazda Original Oil Ultra DPF 5W-30	ACEA C3 0W-30, 5W-30

Engine oil [Australian specs.]

Recommended engine oil
ACEA C1 or JASO DL-1 5W-30

Engine oil [General (R.H.D.), Thailand specs.]

Recommended engine oil
ACEA C3 0W-30, 5W-30